

Problem 1 (6.60). Let D be the cube where x, y, z each vary from -2 to 2 .

- (a) Explain by a symmetry argument why $\int_D xz^2 dV = 0$.
- (b) Evaluate $\int_D (8x + 2xz^2 - 4y^2z + 10) dV$.

Problem 2 (7.6). Let $F(x, y) = \frac{(-y, x)}{x^2 + y^2}$, and let C_R be the circle of radius R centered at the origin and traversed in the counterclockwise direction. Use the parametrization $X(t) = (R \cos t, R \sin t)$, $0 \leq t \leq 2\pi$ to calculate

$$\int_{C_R} F \cdot T ds$$

Problem 3 (7.7). Let C be the triangle with vertices $(0, 0)$, $(1, 0)$, $(1, 1)$, traversed counterclockwise. Evaluate the line integral

$$\int_C y^2 dx + x dy$$

Problem 4 (7.35). Let $X(u, v) = (\sqrt{2}uv, u^2, v^2)$, $(1 \leq u \leq 2, 1 \leq v \leq 2)$

- (a) Calculate X_u , X_v and $X_u \times X_v$
- (b) Show that X parametrizes a surface S in $\mathbb{R}^3$, and at each point (x, y, z) of S , $x^2 = 2yz$.
- (c) Find the area of S .
- (d) Calculate the integral $\int_S y d\sigma$